CSCI-4041
Spring 2026
Paper Homework 6 - Chapter 14

Name: Vector Eramvolkanosin **Section:** _____

1. Consider the following "greedy" strategy to the rod-cutting problem.

   Greedy Strategy: Define the density of a rod of length i to be p_i/i, that is, its value per inch. The greedy strategy for a rod of length n cuts off a first piece of length i, where $1 \leq i \leq n$, having maximum density. It then continues by applying the greedy strategy to the remaining piece of length $n - i$.

   (a) **(2 pts)** Show, by means of a counterexample, that the above "greedy" strategy does not always determine an optimal way to cut rods.

   density $= \frac{p(i)}{i}$

   $1 \leq i \leq n$

i	1	2	3	4	5
$p(i)$	2	4	7	8	10
price	2	2.1	2.3	2	2

   (b) **(4 pts)** Computation of your example via the greedy strategy

   $\frac{p(1)}{1} = 2$

   $\frac{p(2)}{2} = 2$

   $\frac{p(3)}{3} = 2.3$

   $\frac{p(4)}{4} = 3$

   $\frac{p(5)}{5} = 2$

   length $= 5$

   $5 - 3 = 2$

   total cost $= 2.3$

   $2 - 2 = 0$

   total cost $= 2.3 + 2.1$

   * takes maximum

   $= 4.4$

   (c) **(4 pts)** Correct Solution (by any strategy)

   Just do $\rightarrow p(1) + p(1) + p(1) + p(1) + p(1) = 10$

   - price is greater than $p(3) + p(2)$

2. Use the substitution method to show the solution to the following recurrence is $\Omega(2^n)$.

$$P(1) = 1$$

$$P(n) = 1 + \sum_{k=1}^{n-1} P(k)P(n-k)$$

(a) **(4 pts)** Assumption / Hypothesis

$P(n) \geq (c)(2^n)$ $\quad$ $P(k) \geq (\frac{1}{4})\, 2^k$

where $c > 0$ $\rightarrow c = \frac{1}{4}$ $\quad$ for $k > n$

and $n \geq 1$

(b) **(6 pts)** Proof

* base case

$n = 1$, $P(n) = 1$, $(\frac{1}{4}) \cdot 2^n = 0.5 \rightarrow 1 > 0.5$

$n = 2$, $P(n) = 1 + (1)(1) = 2$, $(\frac{1}{4}) 2^n = 1 \rightarrow 2 > 1$

$n = 3$, $P(n) = 1 + (1)(2) + (2)(1) = 5$, $(\frac{1}{4}) 2^n = 2 \rightarrow 5 > 2$

$n = 4$, $P(n) = 1 + (1)(5) + (2)(2) + (5)(1) = 15$, $(\frac{1}{4}) 2^n = 4$, $15 > 4$

* if $P(k) \geq (\frac{1}{4}) 2^k$

$$1 + \sum_{k=1}^{n-1} P(k)P(n-k) \geq 1 + \sum_{k=1}^{n-1} \frac{1}{4}(2^k) \cdot (\frac{1}{4}) \cdot (2^{n-k})$$

$$= \frac{(n-1)}{16}(2^n) \geq \frac{1}{4}(2^n)$$

$$= \frac{(n-1)}{16} \geq (\frac{1}{4}) \rightarrow n \geq 5$$

true for $n \geq 5$ so

$P(n) = \Omega(2^n)$

3. Consider the matrix-chain multiplication problem with an input chain of length n.

(a) **(4 pts)** Describe/sketch the subproblem graph for the dynamic programming algorithm given in the textbook.

- there will be one vertex for each subproblem $m[i,j]$
  ↳ the subproblems in this case are $m[i,j]$
- there is an edge from $m[i,j]$ to every subproblem

(b) **(3 pts)** How many vertices does it have?

- there are one vertex per $[i,j]$
  where $1 \leq i \leq j < n$

$$\binom{n}{2} + n = \frac{n(n+1)}{2}$$

(c) **(3 pts)** How many edges does it have?

- for each subproblem $m[i,j]$ it produces $2(j-i)$ edges

$$= \frac{n(n^2-1)}{3}$$

4. Consider the antithetical variant of the matrix-chain multiplication problem where the goal is to parenthesize the sequence of matrices so as to maximize, rather than minimize, the number of scalar multiplications. Does this problem exhibit optimal substructure?

(a) **(5 pts)** State the recurrence relation between a given problem and the related sub-problems.

$$m[i,j] = \begin{cases} 0 & \text{if } i=j \\ \min\{m[i,k] + m[k+1,j] + p_{i-1}p_k p_j & \text{if } i<j \end{cases}$$

- a recursive equation that assumes the value of k is known
- all value of k is tried, since the optimal parenthesization is one of the values of k
- k values must all be checked

(b) **(5 pts)** Give some explanation or sketch a proof as to why or why not.

- a recursive equation that assumes the value of k is known
- all value of k is tried, since the optimal parenthesization is one of the values of k
- k values must all be checked
- the m[i,j] gives the costs of optimal solutions to subproblems

5. Determine an LCS of $X = [1, 0, 0, 1, 0, 1, 0, 1]$ and $Y = [0, 1, 0, 1, 1, 0, 1, 1, 0]$

(a) **(8 pts)** Table of memo-ized solutions to sub-problems.

i/j	0	1	2	3	4	5	6	7	8	9
0	0	0	0	0	0	0	0	0	0	0
1	0	0	1	1	1	1	1	1	1	1
2	0	1	1	2	2	2	2	2	2	2
3	0	1	1	2	2	2	3	3	3	3
4	0	1	2	2	3	3	3	4	4	4
5	0	1	2	3	3	3	4	4	4	5
6	0	1	2	3	4	4	4	5	5	5
7	0	1	2	3	4	4	5	5	5	6
8	0	1	2	3	4	5	5	6	6	6

(b) **(2 pts)** Reconstructed Solution (a common subsequence of max length)

$c[8,9] \to 1 \neq 0$

$c[7,9] \to 0 = 0$

$c[6,8] \to 1 = 1$

$c[5,7] \to 0 \neq 1$

$c[4,7] \to 1 = 1$

$c[3,6] \to 0 = 0$

$c[2,5] \to 0 \neq 1$

$c[2,4] \to 0 \neq 1$

$c[2,3] \to 0 = 0$

$c[1,2] \to 1 = 1$